Lab 6 Submission Package

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Upload the four labeled sections below according to the form fields.

Submission checklist

1. UPLOAD: Coverage (Code and Branch) Analysis - Coverage_Code_and_Branch_Analysis_155.png

2. UPLOAD: CodeCoverageTest_xxxx_application.py - CodeCoverageTest_155_loan_eligibility.py

3. UPLOAD: BranchCoverageTest_xxxx_application.py - BranchCoverageTest_155_loan_eligibility.py

4. UPLOAD: Minimum 2 commits evidence - two_required_commits_screenshot.png

1. UPLOAD: Coverage (Code and Branch) Analysis

File: Coverage_Code_and_Branch_Analysis_155.png
Submit in: Upload this image in the form field asking for terminal output screenshots.

Code Coverage Analysis

```
============================= test session starts =============================
platform win32 -- Python 3.13.0, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 6
plugins: platformdirs-4.13.0
collected 9 items

CodeCoverageTest_155_loan_eligibility.py .........                       [100%]

============================== 9 passed in 0.19s ==============================
Name                                       Stmts   Miss  Cover   Missing
------------------------------------------------------------------------
CodeCoverageTest_155_loan_eligibility.py      29      0   100%
loan_eligibility.py                           33      0   100%
------------------------------------------------------------------------
TOTAL                                         62      0   100%
```

Branch Coverage Analysis

```
============================= test session starts =============================
platform win32 -- Python 3.13.0, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 6
plugins: platformdirs-4.13.0
collected 22 items

BranchCoverageTest_155_loan_eligibility.py ......................        [100%]

============================== 22 passed in 0.18s ==============================
Name                  Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------------------
loan_eligibility.py      33      0     24      0   100%
-------------------------------------------------------------------
TOTAL                    33      0     24      0   100%
```

2. UPLOAD: CodeCoverageTest_xxxx_application.py

File: CodeCoverageTest_155_loan_eligibility.py
Submit in: Upload this source file in the code coverage test cases field.

```
import pytest

from loan_eligibility import credit_band, interest_rate, is_eligible


def test_credit_band_returns_each_band():
    assert credit_band(800) == "excellent"
    assert credit_band(700) == "good"
    assert credit_band(600) == "fair"
    assert credit_band(500) == "poor"


def test_credit_band_rejects_scores_outside_valid_range():
    with pytest.raises(ValueError, match="invalid score"):
        credit_band(299)

    with pytest.raises(ValueError, match="invalid score"):
        credit_band(901)


def test_eligible_applicant():
    assert is_eligible(30, 50000, "salaried", 800) is True


def test_ineligible_applicant_for_age():
    assert is_eligible(20, 50000, "salaried", 800) is False


def test_ineligible_applicant_for_income():
    assert is_eligible(30, 24999, "salaried", 800) is False


def test_ineligible_applicant_for_poor_credit():
    assert is_eligible(30, 50000, "salaried", 500) is False


def test_unknown_employment_is_rejected():
    with pytest.raises(ValueError, match="unknown employment"):
        is_eligible(30, 50000, "contractor", 800)


def test_interest_rate_for_each_credit_band():
    assert interest_rate(800, "salaried", False) == 8
    assert interest_rate(700, "salaried", False) == 10
    assert interest_rate(600, "salaried", False) == 13


def test_interest_rate_adjustments():
    assert interest_rate(800, "self_employed", True) == 7
```

3. UPLOAD: BranchCoverageTest_xxxx_application.py

File: BranchCoverageTest_155_loan_eligibility.py
Submit in: Upload this source file in the branch coverage test cases field.

```
import pytest

from loan_eligibility import credit_band, interest_rate, is_eligible


@pytest.mark.parametrize(
    ("score", "expected"),
    [
        (299, "invalid"),
        (300, "poor"),
        (549, "poor"),
        (550, "fair"),
        (649, "fair"),
        (650, "good"),
        (749, "good"),
        (750, "excellent"),
        (900, "excellent"),
        (901, "invalid"),
    ],
)
def test_credit_band_boundaries(score, expected):
    if expected == "invalid":
        with pytest.raises(ValueError, match="invalid score"):
            credit_band(score)
    else:
        assert credit_band(score) == expected


@pytest.mark.parametrize(
    ("age", "income", "employment", "score", "expected"),
    [
        (20, 50000, "salaried", 800, False),
        (61, 50000, "salaried", 800, False),
        (21, 24999, "salaried", 800, False),
        (60, 39999, "self_employed", 800, False),
        (21, 25000, "salaried", 550, True),
        (60, 40000, "self_employed", 650, True),
        (30, 50000, "salaried", 549, False),
    ],
)
def test_eligibility_decision_branches(
    age, income, employment, score, expected
):
    assert is_eligible(age, income, employment, score) is expected


def test_eligibility_rejects_unknown_employment():
    with pytest.raises(ValueError, match="unknown employment"):
        is_eligible(30, 50000, "contractor", 800)


@pytest.mark.parametrize(
    ("score", "employment", "has_collateral", "expected"),
    [
        (800, "salaried", False, 8),
        (700, "salaried", True, 8),
        (600, "self_employed", False, 14),
        (500, "self_employed", True, 12),
    ],
)
def test_interest_rate_adjustment_branches(
    score, employment, has_collateral, expected
):
    assert interest_rate(score, employment, has_collateral) == expected
```

4. UPLOAD: Minimum 2 commits evidence

File: two_required_commits_screenshot.png
Submit in: Upload this image in the minimum two commits field.

```
> git log b861b9e d3d473d --format="commit, author, date, message"
commit d3d473d
Author: Dhanush V Biradar <dhanush.v.biradar4@gmail.com>
Date:   2026-11-04

    Add branch coverage tests

commit b861b9e
Author: Dhanush V Biradar <dhanush.v.biradar4@gmail.com>
Date:   2026-11-04

    Add code coverage tests

commit a084d54
Author: Anusha Balamurali <anushaba0@gmail.com>
Date:   2026-10-27

    Add Lab 6 loan eligibility
```